SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1709**

CO1-AT2-Constraint-Based Problem Solving

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 20%

QUESTIONS:3 **Total Marks:30**

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

   Constraints:

   i. D1 cannot work Night shift

   ii. D2 must work before D3 (Morning < Afternoon < Night)

   iii. Only one doctor per shift

   iv. D3 cannot work Morning

   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right

   ii. Cost of each move = 1

   iii. Cannot pass through obstacles

   iv. Use Manhattan Distance heuristic

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost**.**

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.
   The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

   Constraints:

   - The robot can move up, down, left, right

- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:
a) Analyze all the given constraints and explain how they affect the robot's decision-making process.
b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

Code of Conduct:

I G Pavan kumar / (192212293) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Shaik Mahaboob Basha

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Problem Understanding	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly	6
Constraint Analysis	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization	6
Solution Development	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints	7.5

Application of Concepts	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts	6
Justification	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided	4.5

1. Backtracking Search for Doctor Shift Assignment

Aim:

To apply the Backtracking Search algorithm to assign doctors to hospital shifts while satisfying all the given constraints and to obtain a valid schedule.

Problem Statement:

A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night).

Constraints

1. D1 cannot work the Night shift.
2. D2 must work before D3 (Morning < Afternoon < Night).
3. Only one doctor can be assigned to each shift.
4. D3 cannot work the Morning shift.
5. Every doctor must be assigned exactly one shift.

Algorithm (Backtracking Search)

Step 1: Start with no assignments.

Step 2: Assign a shift to the first doctor.

Step 3: Check whether all constraints are satisfied.

Step 4: If the assignment is valid, assign a shift to the next doctor.

Step 5: If any constraint is violated, backtrack and try another shift.

Step 6: Repeat the process until all doctors are assigned valid shifts.

Step 7: Display the final schedule.

Procedure:

1. Consider the doctors D1, D2, and D3.
2. Consider the shifts Morning, Afternoon, and Night.
3. Assign shifts one by one using the Backtracking Search method.
4. After each assignment, verify all the given constraints.
5. If any assignment violates a constraint, return to the previous step and choose another shift.
6. Continue until all doctors receive valid shifts.

Execution:

Step 1: Assign D1

Possible shifts:

- Morning
- Afternoon
- Night (Not allowed)

Choose:

D1 → Morning

Doctor	Shift
D1	Morning
D2	—
D3	—

Constraint Check

- D1 is not assigned Night
- No duplicate shift

Step 2: Assign D2

Remaining shifts:

- Afternoon
- Night

Choose:

D2 → Afternoon

Doctor	Shift
D1	Morning
D2	Afternoon
D3	—

Constraint Check

- D2 is before D3 (possible)
- No duplicate shift

Step 3: Assign D3

Remaining shift:

- Night

Assign:

D3 → Night

Doctor	Shift
D1	Morning
D2	Afternoon
D3	Night

Constraint Check

- D3 is not assigned Morning
- D2 comes before D3
- One doctor per shift
- Every doctor has one shift

Valid assignment found.

Final Schedule

Doctor	Assigned Shift

D1	Morning
D2	Afternoon
D3	Night

Result:

The Backtracking Search algorithm was successfully applied to assign doctors to hospital shifts.

2. Breadth-First Search (BFS) for Robot Path Finding

Aim:

To apply the Breadth-First Search (BFS) algorithm to find the shortest path for a robot in a 5 × 5 grid from the Start (S) position to the Goal (G) while avoiding obstacles.

Problem Statement:

A robot operates in a 5 × 5 grid and must move from Start (S) to Goal (G).

Constraints:

1. Movement allowed: Up, Down, Left, Right
2. Cost of each move = 1
3. The robot cannot move through obstacles.
4. Manhattan Distance heuristic is used to estimate the distance to the goal.

Grid Representation:

S				
		X		
	X	**X**		
	X			
				G

Where:

- **S** = Start (0,0)
- **G** = Goal (4,4)
- **X** = Obstacle

Algorithm (Breadth-First Search)

Step 1: Place the Start node in the queue.

Step 2: Remove the first node from the queue.

Step 3: Check whether it is the Goal.

Step 4: If not, add all valid neighbouring nodes (Up, Down, Left, Right).

Step 5: Mark visited nodes.

Step 6: Repeat until the Goal node is reached.

Step 7: Trace back the path from Goal to Start.

Step-by-Step Node Expansion

Step	**Expanded Node**	**Queue After Expansion**
1	S (0,0)	(0,1), (1,0)
2	(0,1)	(1,0)
3	(1,0)	(2,0)
4	(2,0)	(2,1)
5	(2,1)	(3,1), (2,2)
6	(3,1)	(2,2), (4,1), (3,2)
7	(2,2)	(4,1), (3,2)
8	(4,1)	(3,2), (4,0)
9	(3,2)	(4,0), (3,3)
10	(3,3)	(4,3)
11	(4,3)	(4,4) Goal
12	Goal Reached	Stop

Priority Queue (Open List) Updates

Step	**Queue**
Initial	S
After Step 1	(0,1), (1,0)
After Step 2	(1,0)
After Step 3	(2,0)
After Step 4	(2,1)
After Step 5	(3,1), (2,2)
After Step 6	(2,2), (4,1), (3,2)
After Step 7	(4,1), (3,2)
After Step 8	(3,2), (4,0)
After Step 9	(4,0), (3,3)

After Step 10	(4,3)
After Step 11	Goal

Optimal Path

S

↓

(1,0)

↓

(2,0)

→

(2,1)

↓

(3,1)

→

(3,2)

→

(3,3)

↓

(4,3)

→

G

Path on Grid

S		X		
	S	X		
			X	
X				
		X		G

Cost Calculation

Each move costs **1**.

Total moves:

- (0,0) → (1,0)
- (1,0) → (2,0)
- (2,0) → (2,1)
- (2,1) → (3,1)
- (3,1) → (3,2)

- (3,2) → (3,3)
- (3,3) → (4,3)
- (4,3) → (4,4)

Total Cost = 8

Result

The Breadth-First Search (BFS) algorithm successfully found the shortest path from the Start (S) to the Goal (G) while avoiding obstacles. The robot reached the destination with an optimal path cost of 8, satisfying all the given constraints.

3. Autonomous Rescue Robot Using AI Search Strategy

Aim:

To analyze the problem of an autonomous rescue robot operating in a disaster-affected building and develop an AI-based solution using Uniform Cost Search (UCS) with an Online Search Agent to safely reach the survivor while minimizing the total navigation cost.

Problem Statement:

An autonomous rescue robot is deployed in a disaster-affected building. The robot starts from S (Start) and must reach the survivor at G (Goal).

Constraints:

- The robot can move Up, Down, Left, Right.
- Some rooms are blocked by obstacles.
- The robot can only observe adjacent cells.
- Each movement costs 1.
- Entering risky zones adds an extra cost of 2.
- The robot must minimize the total travel cost.
- The robot updates its knowledge dynamically while moving.

(a) Analysis of Constraints

Constraint	Effect on Decision-Making
Movement only in four directions	The robot can explore only adjacent rooms.
Obstacles cannot be crossed	The robot must find an alternate path around blocked rooms.
Limited sensing capability	The robot has incomplete knowledge and discovers the environment while moving.
Movement cost = 1	Every movement increases the total path cost.
Risky zone additional cost = 2	The robot avoids risky areas whenever possible to reduce total cost.
Dynamic environment	The robot continuously updates its map and changes its path when new information becomes available.

(b) Solution Approach

Search Strategy Used

Uniform Cost Search (UCS) with an Online Search Agent

Steps Involved

Step 1: The robot starts from the initial position S.

Step 2: It observes only the neighbouring cells (limited sensing).

Step 3: It assigns the movement cost:

- Normal cell = 1
- Risky cell = 3 (1 movement + 2 risk cost)

Step 4: Uniform Cost Search selects the path with the lowest cumulative cost.

Step 5: Whenever new obstacles or risky zones are detected, the robot updates its internal map.

Step 6: The robot recalculates the lowest-cost path if necessary.

Step 7: This process continues until the robot reaches the survivor G.

(c) Application of AI Concepts

1. Intelligent Agent

The rescue robot acts as an Intelligent Agent because it:

- Perceives the environment using sensors.
- Processes information.
- Makes decisions.
- Performs actions to achieve the goal.

2. Rationality

The robot behaves rationally by:

- Selecting the safest path.
- Minimizing travel cost.
- Avoiding risky areas whenever possible.
- Updating decisions based on newly discovered information.

3. Environment Type

Property	**Description**
Partially Observable	The robot sees only nearby rooms.
Dynamic	The robot continuously updates its knowledge while moving.
Sequential	Every movement affects future decisions.
Discrete	The building is divided into separate grid cells.
Single Agent	Only one rescue robot performs the task.

(d) Justification of the Chosen Approach

The combination of Uniform Cost Search (UCS) and an Online Search Agent is the most suitable because:

- UCS always finds the minimum-cost path.
- It considers both movement cost and additional risky-zone cost.
- The Online Search Agent can make decisions even without complete knowledge of the environment.
- It dynamically updates the path whenever new obstacles or risky areas are discovered.
- This approach ensures both safe navigation and optimal rescue operations.

Advantages

- Finds the least-cost path.
- Handles risky areas effectively.
- Adapts to unknown environments.
- Suitable for real-time rescue missions.
- Improves navigation safety.

Result

The autonomous rescue robot successfully applies Uniform Cost Search (UCS) together with an Online Search Agent to navigate through a partially observable environment.